

Material Composition and Supply Chain Analysis of Auxiliary Electric Motors in Battery Electric Vehicles

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Executive Summary

This report provides a detailed analysis of the material composition, weight, specific alloys, cost structures, and supply chain dynamics for three categories of auxiliary electric motors integral to the functioning of modern Battery Electric Vehicles (BEVs): Small Stepper Motors (<50W), Medium DC Motors (50-500W), and Medium Stepper Motors (50-200W). The transition to electric mobility has placed a significant emphasis not only on the main traction motor but also on the numerous smaller motors that power essential vehicle functions, from climate control and power steering to automated seat adjustments and advanced driver-assistance systems. Our analysis reveals a complex interplay between material science, manufacturing cost, and geopolitical supply chain risks. Key materials such as copper, electrical steel, aluminum, plastics, and particularly permanent magnets, form the foundation of these motors. The choice between materials like lower-cost ferrite magnets and high-performance Neodymium Iron Boron (NdFeB) rare earth magnets dictates a motor's efficiency, size, and cost [60, 61, 65]. The report highlights that the cost of these motors is heavily influenced by the volatile prices of raw materials, especially copper and rare earth elements [74, 97, 98]. Furthermore, the analysis underscores a critical vulnerability in the automotive sector's supply chain: a profound dependence on China for the mining, processing, and production of rare earth magnets [81]. In response, global automakers and governments are actively pursuing strategies including supply chain diversification, vertical integration through upstream investments, and research into alternative motor technologies to mitigate these risks and ensure a stable supply of these critical components for the future of electric mobility [81].

1.0 Small Stepper Motors (<50W)

1.1 General Characteristics and Applications

Small stepper motors, with a power rating typically below 50 watts, are fundamental electromechanical components in BEVs, engineered for precise, discrete rotational movements rather than continuous rotation [1, 4, 5]. Their ability to move in exact increments, or "steps," makes them indispensable for a wide array of automated functions that require high accuracy and repeatability [1, 3]. Within the automotive context, these motors are deployed in systems such as dashboard instrument gauges for speedometers and tachometers, climate control (HVAC) systems for directing airflow through vents, adaptive headlight leveling and positioning systems, and automated adjustments for mirrors and seats [3, 6]. More advanced applications can include precise control over fuel injection, engine idle speed, and automatic transmission valves in hybrid vehicles [6]. The operation of these motors is governed by an external driver circuit or microcontroller, which energizes the stator coils in a specific sequence to produce the desired step-by-step motion [1, 4]. The reliability and precision of these motors are paramount, as they must function flawlessly across a wide range of temperatures and environmental conditions encountered in a vehicle.

1.2 Material Composition and Weight

The material composition of small stepper motors is centered around a stationary stator and a rotating rotor, with the specific materials varying significantly based on the motor's design classification. The stator universally contains windings made from high-conductivity copper wire [4]. When energized, these coils generate the electromagnetic fields that drive the motor's rotation. The housing and structural components may incorporate plastics and aluminum for weight reduction and protection.

There are three primary types of stepper motors, distinguished mainly by their rotor construction [1, 3, 4]. The **Permanent Magnet (PM)** stepper motor utilizes a rotor that is a permanent magnet, often a magnetized cylinder made of high-retentive steel with defined north and south poles [1, 3, 4]. The interaction between this permanent magnet rotor and the electromagnetic fields of the stator coils causes rotation. **Variable Reluctance (VR)** stepper motors feature a non-magnetized rotor constructed from a soft magnetic material, typically laminated soft iron, with teeth machined onto its surface [1, 3, 4, 5]. The rotor moves by aligning its teeth with the energized stator poles, following the path of least magnetic reluctance. This design is simple and robust but lacks holding torque when unpowered. The most common and advanced type found in demanding applications is the **Hybrid Synchronous Stepper Motor**. This design combines features of both PM and VR types, featuring a complex rotor with a permanently magnetized core, often using rare earth elements, covered by two soft iron "cups" [1, 3, 4, 5]. These cups have fine teeth machined on their surface, creating a highly efficient and precise magnetic coupling with the stator.

The weight of these small motors is a function of their size and material density. While specific data for motors under 50W is not detailed, an analysis of slightly larger NEMA 17 stepper motors (43.18 x 43.18 mm faceplate), which can operate in or near this power range, provides a useful proxy [7, 8, 9]. These motors typically weigh between 0.2 kg and 0.65 kg [7, 8, 10, 11]. The weight is primarily influenced by the motor's length, or stack height, which corresponds to the amount of copper and steel used, directly impacting its torque output. For the <50W category, weights would likely fall in the lower end of this range, estimated to be between 0.2 kg and 0.4 kg, depending on whether it is a simple PM design or a more material-intensive hybrid design [7, 8].

1.3 Specific Alloys and Magnet Grades

The performance and efficiency of small stepper motors are critically dependent on the selection of specialized alloys and magnet grades. For the stator core, which must efficiently channel magnetic flux, manufacturers almost exclusively use **non-grain oriented electrical steel (NOES)** [21, 22, 23, 25]. Unlike grain-oriented steel used in transformers, NOES is isotropic, meaning it exhibits similar magnetic properties in all directions [22, 23, 25]. This is essential for the rotating magnetic fields found in motors. Common grades include M19, M27, and M36 silicon steel [25]. M19 is a versatile low-carbon silicon steel suitable for lower-frequency applications [25]. M27 offers higher magnetic permeability and lower core losses, making it better for high-frequency operation [25]. M36, with a higher silicon content, provides even lower core loss, prioritizing energy efficiency [25]. The thickness of the steel laminations is also a critical design parameter. For automotive drives, thicknesses typically range from 0.20 mm to 0.35 mm [21]. Thinner gauges are crucial for minimizing eddy current losses, which are proportional to the square of the thickness and become a major source of inefficiency at higher operating frequencies [22].

For permanent magnet and hybrid stepper motors, the choice of magnet material is paramount. While some basic PM steppers may use high-retentive steel, high-performance hybrid steppers rely on sintered **Neodymium Iron Boron (NdFeB)** magnets [28, 30]. These rare earth magnets offer the highest magnetic energy product available, enabling the design of compact motors with high torque density [31, 33]. NdFeB magnets are classified by grades, such as N42, N48SH, or N45UH [29, 31]. The

number (e.g., 42, 48) indicates the maximum energy product in MGOe, a measure of magnetic strength [29, 31]. The letters (e.g., H, SH, UH, EH) denote the magnet's intrinsic coercivity and maximum operating temperature [29, 31]. An 'H' (High Coercivity) grade might be rated for 120°C, while 'SH' (Super High) is rated for 150°C, 'UH' (Ultra High) for 180°C, and 'EH' (Extra High) for 200°C [29]. Given the demanding thermal environment within a vehicle, automotive applications necessitate the use of higher-grade magnets, such as the SH, UH, or EH series, to prevent irreversible demagnetization at elevated operating temperatures [29]. These magnets are susceptible to corrosion and typically require a protective coating, such as Nickel-Copper-Nickel (NiCuNi) or epoxy [28, 31].

1.4 Cost Structure and Key Drivers

The cost structure of small stepper motors is a composite of material costs, manufacturing expenses, labor, and production volume. The primary material cost drivers are copper for the windings, electrical steel for the stator laminations, and, most significantly for hybrid motors, the permanent magnets. Historical data, though dated to 1999, provides insight into the relative cost contributions. For a DC brushless permanent magnet motor, which is technologically analogous to a modern hybrid stepper, the magnet material accounted for over 70% of the total material cost (\$180 out of \$253) [74]. While absolute prices have changed dramatically, this highlights the disproportionate impact of rare earth magnet costs on the final product.

The price of copper is another major variable. For fractional horsepower motors, copper typically constitutes 9-10% of the total weight [13, 15, 16]. As copper prices are subject to significant volatility driven by global supply, demand from the green energy transition, and geopolitical factors, fluctuations can directly impact motor manufacturing costs and profit margins [97, 98, 100]. The cost of electrical steel, while less volatile than copper or rare earths, also contributes, with higher-grade, thinner-gauge steels commanding a premium price.

Beyond raw materials, manufacturing complexity is a key cost driver. The intricate assembly of a hybrid stepper rotor and the precision required for winding the stator coils add to labor and processing costs. Production volume plays a crucial role in amortizing these fixed costs; economies of scale achieved through high-volume, automated production can significantly reduce the cost per unit. Therefore, the final cost of a small stepper motor is a delicate balance between the performance specifications dictating material choice and the economic realities of commodity markets and manufacturing scale.

1.5 Supply Chain Analysis

The supply chain for small stepper motors is global and complex, with a particular vulnerability related to its most critical materials. Major manufacturers of small and micromotors include large, diversified Japanese firms like Nidec Corporation and Mabuchi Motor, as well as Hong Kong-based Johnson Electric [89, 90, 92]. These companies operate global production facilities, often located in Asia, to serve the automotive industry.

The most significant supply chain challenge lies with the rare earth elements (REEs) required for high-performance NdFeB magnets. The global REE supply chain is overwhelmingly dominated by China, which as of 2025, controls approximately 70% of global mining, 85% of refining, and around 90% of the production of finished NdFeB magnets [81]. This near-monopoly creates a critical single-point-of-failure risk for the entire automotive industry, which is increasingly reliant on these magnets for both auxiliary motors and EV traction motors. Geopolitical tensions or changes in Chinese export policies, such as the implementation of export licensing for magnet materials, can lead to immediate price spikes and supply disruptions, as have been experienced by major automakers [81].

In response to this vulnerability, a multi-pronged strategic response is underway across the industry and at governmental levels. Automakers are pursuing **diversification** by seeking out and paying premiums for non-Chinese magnet suppliers, such as Neo Performance Materials' facility in Estonia [81]. Companies are also engaging in **upstream investments** and vertical integration to secure supply. General Motors, for example, has partnered with MP Materials for a U.S.-based mine-to-magnet supply chain [81]. Similarly, USA Rare Earth is developing a domestic supply chain in the United States [82, 83, 85]. Governments are providing support through initiatives like the U.S. and Australia's critical mineral financing packages and Europe's Critical Raw Materials Act, which aim to foster domestic production [81]. Finally, **technology substitution** is being explored, with research focused on developing magnets that use less of the most critical rare earths (like dysprosium) or designing rare-earth-free motor architectures, though NdFeB magnets are expected to remain dominant for performance-critical applications for the foreseeable future [81].

2.0 Medium DC Motors (50-500W)

2.1 General Characteristics and Applications

Medium-power Direct Current (DC) motors, operating in the 50-watt to 500-watt range, are workhorses within the BEV architecture, responsible for powering a multitude of functions that require more substantial torque and power than small stepper motors can provide. These motors are found in applications such as electric coolant pumps, HVAC blower fans, power window lifts, sunroof mechanisms, and motorized seat adjustments [88, 93]. They are broadly categorized into two main types: brushed DC motors and brushless DC (BLDC) motors [35, 38]. Brushed DC motors are a traditional, simpler technology that uses physical carbon brushes to mechanically commutate the current in the windings [35]. BLDC motors, in contrast, use an electronic controller to switch the current, eliminating the wear-prone brushes [38, 40]. This results in higher efficiency, longer service life, lower maintenance, and better speed control, making BLDC motors the preferred choice for many modern BEV applications, despite their higher initial cost and complexity [37, 38].

2.2 Material Composition and Weight

The material composition of medium DC motors is dominated by metals. The motor housing, or casing, is frequently constructed from die-cast **aluminum** to provide a lightweight yet robust enclosure with excellent heat dissipation properties [35, 36, 37]. In some designs, particularly for more rugged applications, **steel** may be used for the housing [53, 55]. The electrical windings are almost universally made from **100% copper wire** due to its superior electrical conductivity, which is essential for efficient operation and managing the higher currents in this power class [35, 37, 48].

The internal magnetic components differ significantly between brushed and BLDC motors. Traditional brushed DC motors often employ **ferrite permanent magnets** in the stator [35]. These ceramic magnets are cost-effective and offer good resistance to corrosion and high temperatures. The rotor contains the copper windings, and commutation is achieved via **carbon brushes** that make physical contact with a commutator [35]. In contrast, a BLDC motor typically has its permanent magnets on the rotor and the copper windings in the stator. While some cost-sensitive BLDC motors may use ferrite magnets, high-performance applications increasingly use stronger **NdFeB rare earth magnets** to achieve higher torque density and efficiency in a more compact form factor [63, 65].

The weight of these motors is substantial, reflecting their higher power output and material content. For example, a 200W DC motor, depending on whether it is a brushed or brushless design and the specific manufacturer, typically weighs between 2.3 kg and 3.6 kg [41, 42, 43, 44, 45]. A 500W motor would be correspondingly heavier. In terms of material breakdown by percentage, DC motors tend to have a relatively high copper content, estimated to be between 15% and 18% of the total weight [13,

15]. The remainder is composed primarily of steel in the magnetic core and shaft, aluminum in the housing, and the permanent magnets, with a smaller fraction of plastics and insulation materials.

2.3 Specific Alloys and Magnet Grades

The selection of alloys for medium DC motors is critical for ensuring structural integrity, magnetic performance, and longevity. For the motor housing, or frame, a variety of materials are used. **Aluminum alloys** are favored for their lightweight properties and excellent thermal conductivity, allowing the housing to act as a heat sink [53, 55]. **Fabricated steel** housings offer a cost-effective and durable alternative, while **cast iron** is reserved for heavy-duty applications where rigidity and vibration damping are paramount [53, 55]. The motor shaft, which transmits the mechanical power, is typically made from high-strength steel. Common grades include plain carbon steels like **SAE 1045** for general use, or higher-strength chrome-molybdenum steels like **4140** for more demanding applications [57, 58].

The choice of permanent magnet material represents a critical design trade-off between cost and performance. **Ferrite magnets**, composed of iron oxide and strontium or barium carbonate, are the low-cost option [60, 62, 64]. They offer excellent resistance to corrosion and high temperatures (operating up to 300°C) and their coercivity (resistance to demagnetization) actually increases with temperature [60, 64]. However, their magnetic field is significantly weaker than that of rare earth magnets. This makes them suitable for cost-sensitive, intermittent-use applications where maximum power density is not the primary goal.

Neodymium (NdFeB) magnets, on the other hand, provide the highest magnetic strength available, allowing for the design of smaller, lighter, and more efficient motors [60, 61, 65]. This is particularly advantageous for BLDC motors in BEVs, where space and weight are at a premium. However, NdFeB magnets are considerably more expensive, more brittle, and more susceptible to corrosion, requiring protective coatings [61, 64, 65]. They also have a lower maximum operating temperature than ferrites. Standard grades can begin to lose magnetism above 80°C, necessitating the use of more expensive high-coercivity grades (e.g., SH, UH, EH series, rated for 150°C to 200°C) for the demanding thermal environment of automotive applications [60, 61].

2.4 Cost Structure and Key Drivers

The cost of medium DC motors is driven by material selection, manufacturing complexity, and production volume. A significant cost differentiator is the choice between a brushed and a brushless design, and within BLDC motors, the choice of magnet type. A traditional brushed DC motor using ferrite magnets is the most cost-effective option. Historical data from 1999 for a conventional DC motor showed a total material cost of approximately \$99 and an estimated OEM cost of \$310, with no specific cost allocated to magnets, implying the use of inexpensive ferrites or an electromagnetic field design [74].

A high-performance BLDC motor using NdFeB magnets is substantially more expensive. The cost of the rare earth magnets alone can be a dominant factor, as seen in analogous motor types [74]. Furthermore, BLDC motors require a complex electronic controller, which adds significantly to the overall system cost. The price of copper is another critical cost driver. With a copper content of 15-18% by weight, these motors are highly sensitive to fluctuations in the copper market [13, 15, 97]. The sustained increase in copper prices, driven by demand from EVs and renewable energy infrastructure, directly translates to higher manufacturing costs for motor producers [97, 99, 100]. The choice of housing material also plays a role, with die-cast aluminum generally being more expensive than fabricated steel. As with smaller motors, achieving high production volumes is essential for reducing per-unit costs by leveraging economies of scale in both material procurement and automated manufacturing processes.

2.5 Supply Chain Analysis

The supply chain for medium DC motors is populated by some of the largest Tier 1 automotive suppliers in the world. Key manufacturers include Robert Bosch GmbH, Denso Corporation, Continental AG, Nidec Corporation, and Johnson Electric [88, 90, 92]. These global giants have extensive manufacturing footprints and long-standing relationships with automakers, supplying a vast portfolio of automotive components, including a wide range of auxiliary motors. Their production facilities are strategically located across North America, Europe, and Asia to serve regional markets efficiently.

The supply chain for the raw materials used in these motors faces distinct challenges. The copper market is characterized by price volatility and a tightening supply-demand balance [98, 99, 101]. Growing demand from global electrification efforts is putting pressure on a supply chain where bringing new mines online is a lengthy and capital-intensive process. This dynamic suggests continued upward pressure on copper prices, directly impacting motor costs.

For high-performance BLDC motors that utilize NdFeB magnets, the supply chain is exposed to the same critical vulnerability discussed for stepper motors: the near-monopoly held by China over the rare earth supply chain [81]. The larger mass of magnetic material required for a 50-500W motor compared to a sub-50W motor makes this dependency even more acute. Any disruption in the availability or pricing of NdFeB magnets from China can have a profound impact on the production of these key BEV components. This reinforces the strategic imperative for automakers and their suppliers to invest in diversifying their sources of rare earth materials and to support the development of alternative, non-Chinese supply chains to ensure long-term resilience.

3.0 Medium Stepper Motors (50-200W)

3.1 General Characteristics and Applications

Medium stepper motors, operating in the 50-watt to 200-watt power range, represent a step up in torque and power from their smaller counterparts while retaining the core benefit of precise, step-based positional control. These motors are typically of the hybrid synchronous design, which offers the best combination of torque, speed, and resolution [68, 69]. A common industry-standard form factor for this class of motor is the NEMA 23, which has a 57mm x 57mm faceplate [68, 69]. In BEVs, these higher-power stepper motors are employed in applications that demand greater force, such as certain types of electronic throttle control, wastegate actuators in turbocharged systems (for hybrid models), and more robust automated valve controls. They may also be used in precision mechatronic systems where a DC motor's continuous rotation is not suitable, but a higher holding torque than a small stepper can provide is necessary. Often, these motors are paired with a gearbox to multiply torque for heavy-duty actuation tasks.

3.2 Material Composition and Weight

The material composition of a medium stepper motor is fundamentally similar to that of a small hybrid stepper, but scaled up in size and mass to handle the increased power and thermal loads. The stator consists of a larger stack of electrical steel laminations and more substantial **copper windings** to carry the higher current [68, 70, 71]. The rotor is a more robust version of the hybrid design, featuring a powerful **permanent magnet** core encased in toothed soft iron sections [68, 69]. The housing is typically a rigid metal structure, often aluminum or steel, designed to manage heat and provide structural support.

For versions equipped with a gearbox, additional materials come into play. The gears themselves can be made from **powder metallurgy**, a cost-effective process for creating complex shapes suitable for

many applications [70]. For high-precision or high-torque gearboxes, the gears are often machined from high-strength **alloy steel** to ensure durability and minimize backlash [71].

The weight of these motors is significantly greater than that of small steppers. While specific weight data for the 50-200W range is not provided in the source material, it can be estimated to be considerably more than the 0.2 kg to 0.65 kg range of the smaller NEMA 17 motors [7, 8, 10, 11]. Given the larger NEMA 23 frame size and the increased volume of copper, steel, and magnetic material required for the higher power rating, a weight range of approximately 0.7 kg to 1.5 kg or more is a reasonable estimate, with geared versions being at the higher end of or exceeding this range.

3.3 Specific Alloys and Magnet Grades

The alloys and magnet grades used in medium stepper motors are selected to maximize performance under higher electrical and mechanical stress. The stator laminations are made from **non-grain oriented electrical steel (NOES)**, with grades like M19, M27, or M36 being common choices [25]. For a motor in this power class, which may operate at higher frequencies and generate more heat, a higher-grade steel like M27 or M36 with a thin gauge (e.g., 0.35mm or less) would be preferred to minimize core losses and improve overall efficiency [21, 22].

The permanent magnet is the heart of a high-torque hybrid stepper motor and is almost exclusively a high-strength **sintered NdFeB magnet** [28, 30]. To produce the required torque and resist demagnetization from the strong magnetic fields and higher operating temperatures, high-coercivity grades are essential. Grades such as **48SH**, **45UH**, or **42EH** are likely candidates [29]. These grades offer not only a strong magnetic field (BHmax of 40-48 MGOe) but also superior thermal stability, with maximum operating temperatures ranging from 150°C to 200°C [29]. This thermal resilience is critical to prevent permanent degradation of the motor's performance in the demanding automotive environment. In geared versions, the choice of **alloy steel** for the gears is crucial for reliability, providing the necessary hardness and fatigue strength to handle the multiplied torque output over the vehicle's lifetime [71].

3.4 Cost Structure and Key Drivers

The cost of medium stepper motors is substantially higher than that of their smaller counterparts, driven directly by the increased quantity and quality of the materials used. The primary cost drivers are the larger mass of copper in the windings, the larger stack of high-grade electrical steel, and, most notably, the larger and higher-grade NdFeB permanent magnet. The cost of the magnet, in particular, increases significantly with both size and thermal grade, making it a dominant component of the overall material bill [29, 74].

If the motor includes a planetary gearbox, this adds another significant layer of cost [70, 71]. The manufacturing of high-precision gears, especially those made from machined alloy steel, is an expensive process [71]. The complexity of assembling the gearbox also adds to labor costs. Therefore, a geared NEMA 23 stepper motor represents a considerable investment compared to a standard, non-geared motor. As with all motor types, manufacturing costs are also influenced by the precision required in machining and assembly, as well as the production volume. The higher performance and torque requirements of these motors necessitate tighter manufacturing tolerances, which further contributes to their higher price point.

3.5 Supply Chain Analysis

The supply chain for medium stepper motors mirrors that of other advanced motor types, with manufacturing dominated by specialized global firms such as Nidec, Johnson Electric, and AMETEK [89, 92]. These companies have the technical expertise to produce the high-precision components required for hybrid stepper motors and their associated gearboxes.

The supply chain for these motors is critically dependent on the availability of high-grade NdFeB magnets [81]. The larger magnet required for a 50-200W motor amplifies the automotive industry's exposure to the risks associated with the China-dominated rare earth supply chain [81]. The procurement of these powerful magnets is a key strategic activity for motor manufacturers, and any volatility in the rare earth market has a direct and significant impact on their production costs and stability. The supply chain for high-precision gearboxes is also specialized, often involving different suppliers than the motor manufacturers themselves. The integration of these two complex components—the motor and the gearbox—requires close coordination between suppliers to ensure compatibility and performance. The overarching supply chain strategy for these motors must therefore address not only the sourcing of raw materials like copper and steel but also the highly concentrated and geopolitically sensitive market for rare earth magnets and the specialized market for precision mechanical components.

4.0 Conclusion: Synthesis of Findings

This comprehensive analysis of auxiliary electric motors in BEVs reveals a sophisticated and critical sub-sector of the automotive industry, governed by a complex balance of material science, economic pressures, and supply chain vulnerabilities. Across small stepper, medium DC, and medium stepper motors, a clear trend emerges: as performance requirements for power, torque, and efficiency increase, so does the reliance on advanced, higher-cost materials. The transition from simple iron rotors to high-grade NdFeB permanent magnets, and from standard steel to thin-gauge, low-loss electrical steel, is central to achieving the compact, powerful designs demanded by modern vehicles [22, 25, 60, 65].

Two materials stand out as the most significant cost drivers and sources of market volatility: **copper** and **rare earth elements** [74, 97]. The price of copper, essential for windings, is subject to upward pressure from the global transition to green energy, directly impacting the base cost of every motor. Even more critical is the cost and availability of NdFeB magnets. For any motor utilizing this technology, the magnet represents a substantial portion of the material cost and exposes the manufacturer to the extreme price volatility and geopolitical risk inherent in the rare earth market [74].

This leads to the report's most critical finding regarding the supply chain: the automotive industry's profound dependence on China for the processing of rare earth elements and the production of permanent magnets constitutes a systemic risk [81]. This single-point-of-failure vulnerability threatens the stability of BEV production worldwide. In response, a strategic shift is underway, with automakers and governments collaborating to de-risk the supply chain through diversification of sourcing, direct investment in non-Chinese mining and processing facilities, and long-term research into alternative materials and motor technologies [81, 82, 84]. The future design, cost, and production of auxiliary motors—and by extension, the advancement of BEVs—will be inextricably linked to the successful management of these material and supply chain challenges.

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